# USA TRIATHLON

## RACE DIRECTOR RESOURCE GUIDE:

## SWIM: WATER CONDITIONS

**INTRODUCTION:**

Before diving too deep into your triathlon planning (pun intended), you will need to identify a body of water that will be able to safely accommodate the swim. Regardless if the swim takes place in a lake, a river, the ocean, or another body of water, it is the part of the race that arguably presents the largest risk. As the Race Director, you create the logistics plans for how athletes enter and exit the water and the number of safety assets positioned along the course. However, the water conditions and how athletes will react to them are almost completely out of your control. Each race presents a different set of circumstances and challenges.

With all of this in mind, it is important to do your homework prior to the event to confirm the swim is set up to be as safe and successful as possible. Every body of water has its own unique characteristics and nuances. The best way to learn the ins and outs of the area is by developing relationships with local groups who interact with the water on a regular basis. They will be able to provide historical information and insight when it comes to recent conditions. Yacht clubs, lifeguards, fishing groups, kayak clubs, surfers, and more can all be great resources to help paint a holistic picture of how the water behaves.

In this section, we will introduce the factors that should be taken into account when first determining the viability of a swim course and re-evaluated annually to ensure conditions remain safe for racing:

- Site Plan & Access Points
- Tides & Currents
- Wind
- Swells
- Underwater Obstructions
- Water Temperature
- Water Quality
- Daylight
- Marine Life
- Commercial Boat Traffic

**Note: While this guide does provide recommendations for the swim portion of the race, all planning and decision-making regarding the event is at the discretion of the Race Director and the agency that governs the body of water. While direction is provided on the competition rules, it is not a replacement for the competition rules. For questions about the rules and their application at your event, please reach out to rulesandofficials@usatriathlon.org.*

## SITE PLAN & ACCESS POINTS:

As you set out to decide if a body of water is suitable to host your swim course, the first thing to look at is the land that surrounds the water. This venue needs to have the square footage and functionality to accommodate all the key components of an event site plan: vehicle access, restrooms, water station, line queues, etc. This is no different than any other checklist you might run through when determining the viability of a venue. Easy peasy.

The piece that gets more complicated is determining the flow of athletes from the land to the water. During your site visit, start by answering these questions about the access points to the water:

| SWIM START | SWIM EXIT | SWIM EXTRACTION | RACE OPERATIONS |
|---|---|---|---|
| Where and how will you safely get athletes in the water? | Where and how will you safely get athletes out of the water? | Where and how will you safely remove athletes unable to complete the swim in emergent and non-emergent situations? | Where and how will you launch safety assets in and out of the water (i.e. kayaks, boats, jetskis, etc) before and after the race? |

*Is this the same physical location or are these two separate locations? If they are the same, how will you structure your course / timeline to prevent crossover?*

Is there unimpeded vehicle access from the extraction point to the nearest hospital (i.e. does not have to cross over the course)

Take note of existing infrastructure that can be used to help facilitate access. Think creatively about how to work with what is available and consider what additional assets can be temporarily installed or brought in to help support the effort. Here are a few common race start/exit methods:

| EXISTING INFRASTRUCTURE | TEMPORARY INFRASTRUCTURE |
|---|---|
| ➔ Beach | ➔ Ramp / staircase |
| ➔ Dock | ➔ Barge |
| ➔ Jetty | ➔ Floating dock |
| | ➔ Boat |

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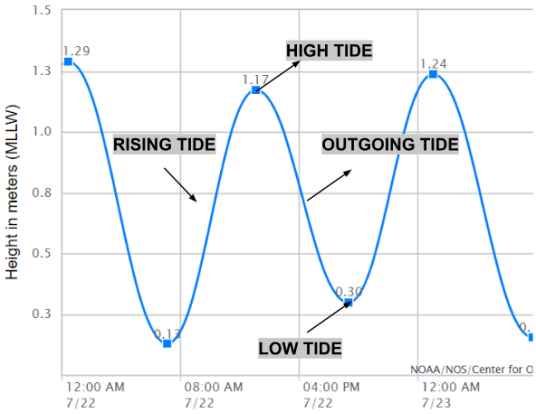
## TIDES & CURRENTS:

If the body of water is affected by tides and/or currents, it's important to ensure that the water conditions are working in your favor on the date and time of your event. To start, let's clarify the difference between a tide and a current:

**Tides:** Tide refers to the rise of and fall of the water levels. Understanding the flow of the tides can shed light on the water's *depth* at a given time. Tides are measured in height, in either feet or meters. They are influenced by the moon's gravitational pull and the sun's position so they follow a predictable

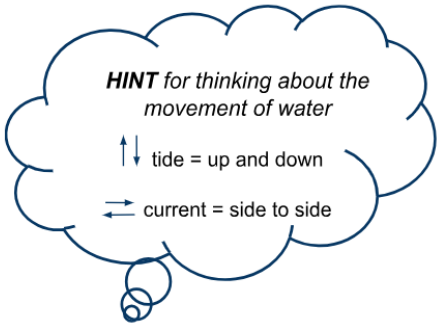
In the context of your swim event, the tides can influence how long you have the necessary depth for athletes to be able to swim. Encountering low tide during the race could mean you quite literally “run out of water.” Tidal changes can impact the viability of the swim course itself and also how athletes will enter/exit the water. For example, in a scenario where athletes are starting from a barge, low water levels may make it unsafe to allow a dive start. In a scenario where the seafloor may be rocky or have debris, losing the water may prevent athletes from swimming all the way to the exit, forcing them to have to walk in unsafe conditions.




**HOW TO READ A TIDE CHART:**

- **Rising Tide**: Once low tide occurs, water levels will start to increase
- **High Tide:** The water level will be at its highest point during high tide
- **Outgoing Tide:** Once high tide occurs, water levels will start to decrease
- **Low Tide:** The water level will be at its lowest during low tide

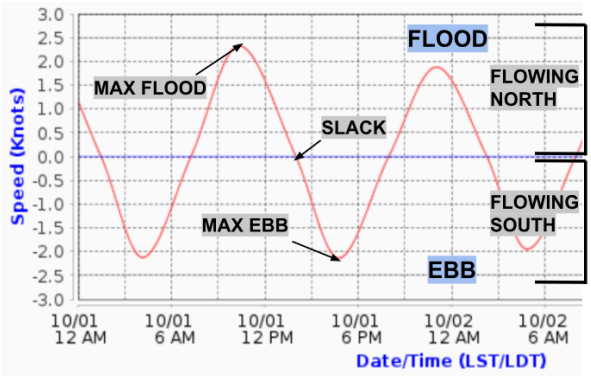
**Currents:** While tides dictate the water depth, currents refer to the horizontal *direction* of the water flow. Currents are described in terms of their speed (measured in knots) and direction. Currents are caused by wind, water density, temperature, and the impacts of the tide. While most of these factors are generally unpredictable, tidal currents can be forecasted since they are dictated by the lunar and solar cycles.




In the context of your swim event, the current impacts the direction and strength of the water flow. In general, the swim should be timed so athletes are swimming with the current as opposed to against. Understanding the point when the current will be at its peak and the point at which it changes direction, may influence how you structure your swim timeline. For example, if the current is going to start weak and increase in strength as the morning progresses, it might be advisable to start your strongest swimmers first and your first timers/non-competitive waves last to provide the weaker swimmers with the strongest current.

**HOW TO READ A TIDAL CURRENT CHART:**

- **Flood:** Rising tides bring flood currents, which move water in the direction of the shore. In this example, the flood current is flowing north
- **Max Flood**: The strength of the current moving toward the shore (north) is at its peak speed
- **Ebb**: Outgoing tides bring ebb currents, which move water away from the shore. In this example, the ebb current is flowing south
- **Max Ebb**: The strength of the current moving away from the shore (south) is at its peak




- **Slack:** Occurs when the tidal current changes direction from flowing in one direction to flowing in the opposite direction. This moment of transition, when the water is nearly still, is the slack tide.

| ***Pro Tip:*** |
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| *The NOAA (National Oceanic and Atmospheric Association) provides charts for predicting tides and tidal currents at specific locations, dates, and times:*<br>• *> Tide Predictions*<br>• *> Tidal Current Predictions* |

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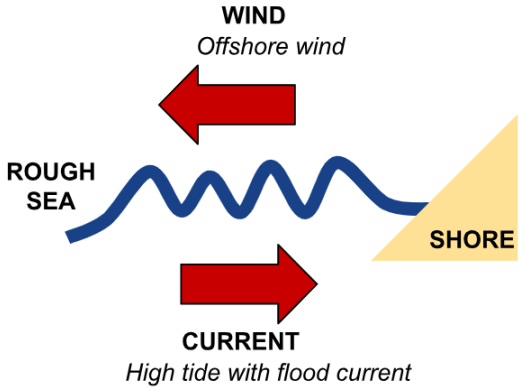
**WIND:**

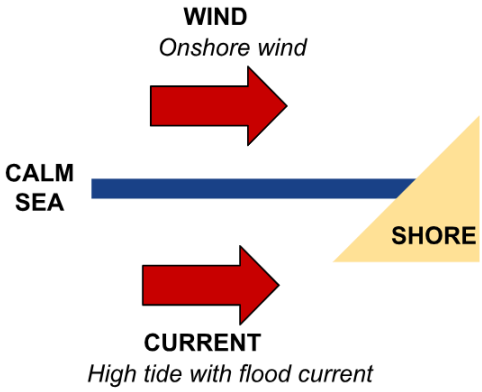
Wind is the flow of air from areas of high pressure to areas of low pressure resulting from changes in temperature. Wind speed and direction influence water conditions, as these factors affect the roughness of the water and swings in temperature. There are two primary types of winds to consider - offshore winds and onshore winds:

| **WIND DIRECTIONS** | | |
|---|---|---|
| | **OFFSHORE WINDS** | **ONSHORE WINDS** |
| **DIRECTION** | Blow from the land toward the sea. If you are standing on the beach facing the ocean, the wind is blowing at your back | Blow from the sea towards the land. If you are standing on the beach facing the ocean, the wind is blowing at your face |
| **WATER** | • During periods of offshore wind, the water tends to be calmer near the shore. This is because wind needs to blow over a large area of water in order to make waves. | • The onshore winds push the waves toward the shore, breaking closer to the beach.<br>• The waves tend to be on the smaller side, but choppy. The messy nature of |

| | | |
|---|---|---|
| | • As you get further out, the wind gets up under the water forming the quintessential, tidy wave that peel mechanically. These waves are larger, which means a bigger crash when they hit the surface. While these waves are ideal for surfing, they can be disruptive for a swim event. | the waves makes them challenging to predict, as they are not as clean as offshore waves.<br><br>• The increased turbulence in the water might make sighting landmarks/buoys more difficult for swimmers. |
| **WATER TEMPERATURE** | • The water temperature tends to decrease as the warmer surface water is carried out to sea. The crashing of larger waves also churns up deeper water which tends to be colder than water just below the surface. | • The water temperature tends to increase as the warm surface water is pushed toward the shore. |

The way the wind interacts with the current also impacts the roughness of the water. As a general rule of thumb, when the wind blows against the current, the waves become steeper and break, but when the tide and the wind are moving in the same direction, the waves tend to flatten and lengthen, creating calmer waters.

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**SWELLS:**

Unlike wind waves, which are created by local winds near the surface of the water, swells are generated by strong and sustained winds initiated by distant storms. Swells can travel thousands of miles across open water and often pick up momentum as they go. Since these storms occur in the distance, there is lead time to be able to track the progression and movement of swells days before they will affect local waters.

**Types of Swells:** There are three primary types of swells:

| GROUNDSWELL | WINDSWELL | HURRICANE SWELL |
|---|---|---|
| A swell that has traveled further than 500 miles. Groundswells start building wherever wintertime is happening in that hemisphere. Tend to be fairly powerful. | A swell that has traveled less than 500 miles and tend to be smaller, weaker, and last for a shorter period of time than groundswells. | A swell that is formed by the strong winds that originate as part of a hurricane or tropical storm. The closer the coastline is to the storm's path, the more powerful the swells will be. |

**Direction:** When projecting the influence of swells, it is important to understand the direction the swell is coming from and the direction the beach is facing. For example, if the swell is coming from the northwest, beaches that are facing north and west will be impacted by the swell more than beaches facing south. This doesn't mean that a beach with a southern orientation will be clear of the swell, just that the waves will be smaller since the beach is facing the opposite direction.

**Landscape of the Coastline:** Natural and man-made barriers can play a crucial role in blocking off beaches from swells and tempering wave strength. These structures or protrusions can help to insulate certain areas from bearing the full force of the water's conditions, which tends to make these sections safer for swimming.

| STRUCTURE TYPE | PHOTO EXAMPLE |
|---|---|
| **Jetty:**<br>A jetty is a long, narrow, man-made structure built to modify the flow of water to protect the coastal areas. These are commonly found at the entrances of harbors and ports, as they help to maintain a calmer channel for ships to enter and exit. | |
| **Barrier Island:**<br>Barrier islands are long, narrow landforms that run parallel to the mainland coast. These islands provide a protective barrier from open ocean swells, allowing for the development of calmer waters in the area between the island and the coast. | |
| **Sandbar:**<br>Submerged or partially submerged sandbanks located offshore or parallel to the shoreline. Sandbars can help to decrease the strength of incoming swells as they cause the waves to break earlier, which reduces their intensity before reaching the beach. | |

**Headlands / Cliffs:**
Headlands are rocky protrusions that jut out into the sea, while cliffs are steep rock formations along the coastline. Both act as natural barriers, blocking the impact of swells on the beaches located behind them. The waves tend to wrap around the headlands or cliffs, resulting in more sheltered and protected beach areas.

***Pro Tip:***

*Surfers love swells! There are a lot of resources that have been developed for surfers to forecast how swells will affect the water and the projected surf:*

*> Surfline (Magic Seaweed)*
*> Swell Info*

*Surfers seek:*

- *Strong offshore winds*
- *Groundswells*
- *Same beach orientation as the swell direction*
- *Open beach area with no coastal structures/protrusions*

*Race Directors seek:*

- *Weak onshore winds*
- *No swells or windswells*
- *Opposite beach orientation as swell direction*
- *Coastal structures/protrusions that provide insulate the swimming area*

*Generally speaking, the beach conditions surfers are hoping for are usually the conditions Race Directors are looking to avoid for their events. If surfers are flocking to the area where your swim is being held, this might be a red flag.*

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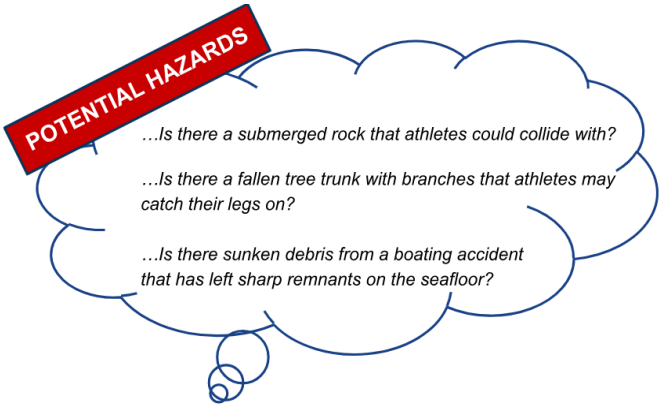
**UNDERWATER OBSTRUCTIONS:**

While the coastline features of a body of water are clearly visible when scouting a location for your swim course, the area below the surface is a bit more of a mystery. The list of potential hazards is endless and until you are able to rule out the existence of dangers below the surface, you should not assume the swim course is safe.




SONAR (Sound Navigation And Ranging) technology can be used for underwater mapping. It emits sound waves that travel through the water

and as they encounter objects, some of the waves are reflected back to the SONAR system indicating the distance, size, and shape of underwater objects. It can also help to provide accurate measurements on the depth of the water in areas along the course.

If SONAR is going to be used prior to the event to rule out any obstructions, schedule the course ride at a time when the tides mirror how they will be during the swim to provide the best representation of race day conditions. During the event, SONAR is also useful to have on-hand as a search and rescue tool to help locate a submerged athlete.

| ***Pro Tip:*** |
| --- |
| *SONAR equipment can be expensive to purchase. Engage with your on-water municipality contacts as police department, fire department, and US Coast Guard vessels are often outfitted with this technology and may be able to provide this service. Also reach out to members of the local boating community, yacht clubs, and fishing groups as there may be individuals who have this technology and can assist leading up to and during the event.* |

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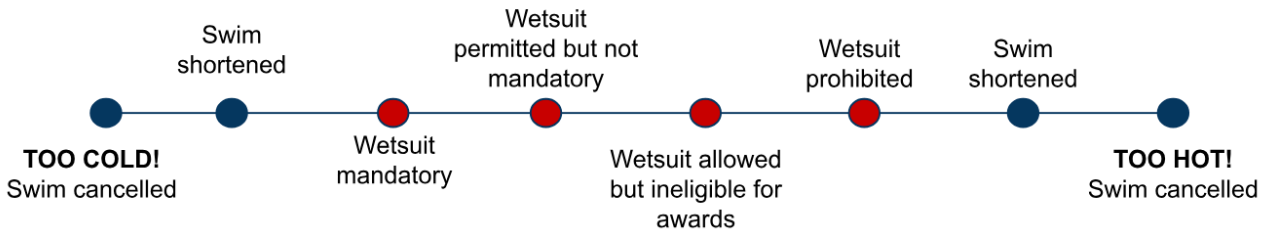
**WATER TEMPERATURE:**

Water temperature is a spectrum. Once temperatures drop below or exceed certain levels, actions are required to maintain athlete safety. These may include course shortening/cancellation measures and/or enacting wetsuit policies. For more information on the specific USAT temperature thresholds, check out the Water Temperature section of the Weather Action Planning Resource Guide. The Official USAT Multisport Competition Rules can also be downloaded at this link.




When conducting research about your body of water, obtain data on the annual temperature readings in the area where your swim is held. To get the best representation of race day temperature readings, keep the following conditions consistent:

- **Location:** USAT recommends the temperature be taken at three points, including the middle of the swim course.

- **Depth:** Temperature readings taken deep below the surface, tend to have lower temperature than those taken at the surface. Official USAT water temperatures are to be taken at 24” below the surface.

- **Time of Day**: In some areas, there can be a large swing in water temperature throughout the day and in others, it may remain fairly consistent. This depends on the position and strength of the sun and fluctuations in air temperature throughout the day. Gather enough data to understand the average swing in temperatures to help prevent any surprises on race day.

- **Tides:** Depending on your body of water, it is possible changes in tide can influence the water temperature. For example, in the San Francisco Bay, the ocean is the source of high tide and ocean water is colder than the bay water. As a result, the water tends to be cooler during high tide and warmer when the tide is low.

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**WATER QUALITY:**

To be a USAT sanctioned event, water testing is a requirement, even for pool swims. USAT recommends Race Directors follow the guidelines established by the agency that governs the body of water, such as the county, city, EPA, pool management, etc. USAT will accept whatever criteria the permitting agency has established for swimability and human contact.

A body of water’s levels of pollutants, bacteria, and acidity impact how athletes will react to prolonged exposure. Once levels increase above certain thresholds, the water quality presents a potential health risk to athletes and swimming-related illnesses become a potential threat. For more information on the primary influences on water quality, the sampling process, and results threshold references, visit the **Swim - Water Quality Analysis Resource Guide**.

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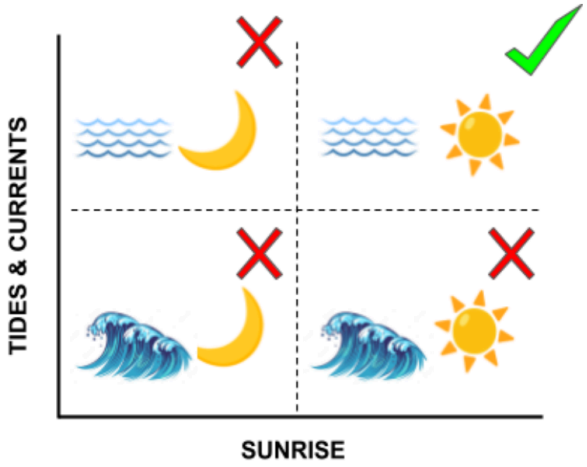
**DAYLIGHT:**

For swims impacted by tides and currents, the window of time when the swim can safely take place is often limited and precise. Another crucial piece for determining the swim timeline is factoring in when the sun rises (for morning events) and when the sun sets (for evening events). In order for on-water safety assets to be able to monitor swimmers throughout the event, there needs to be ample daylight.




While daylight is a crucial component, keep in mind that it can also bring its own set of challenges. The position of

the sun at the time when swimmers will be in the water may affect their ability to sight buoys. Along those lines, if swim safety personnel are looking directly into the sun, it may also impact their ability to observe swimmers.

In short, when it comes to establishing the start time of your swim, the alignment of both the tides/currents and the sun go hand-in-hand. You can't have one without the other.

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**MARINE LIFE:**

Each body of water is home to a vast array of marine life. In this context, it's important to remember swim events are a temporary intrusion on a much larger ecosystem. Depending on the location of the swim, it may be necessary to identify the types of sealife in the area, migration patterns, and population rates. Here are a few examples of common sealife threats to keep an eye on:

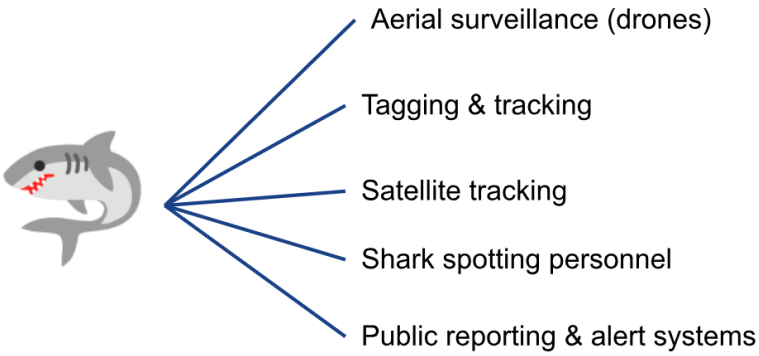
- **Jellyfish:** Some species of jellyfish have stinging tentacles that can cause painful skin irritations or allergic reactions. Jellyfish blooms typically occur during certain seasons and may be prone to some areas more than others based on water temperatures and tides and currents

**DID YOU KNOW?**

If there is any possibility a jellyfish sting could occur during the swim, plan to have vinegar on-hand for the medical team to be able to treat afflicted swimmers.

- **Crocodiles & Alligators:** Typically found in freshwater rivers, lakes, and swamps, alligators and crocodiles can pose a threat to swim events. If your event takes place in a body of water where they are known to be present, consult with local authorities and experts to advise on their locations and behaviors prior to the event.

- **Sharks:** The presence of sharks has become a more prevalent issue in recent years. As waters warm, sharks are beginning to change their migration and routes in order to stay in the waters most comfortable for them. This means they are being sighted in areas where they have never been in the past. Chat with your municipality contacts about their involvement in local and national efforts to track shark movements and ensure the safety of swimmers:



**COMMERCIAL BOAT TRAFFIC:**

Heavy commercial boat traffic in the vicinity of the swim course can create potential safety hazards. The wake and turbulence generated by boats can create challenges for swimmers and of course, the possibility of collision between boats and swimmers is a dangerous possibility.

If commercial boat traffic is a potential concern, before deciding to host the swim it is important to confirm completion of the risk mitigation checklist will be possible.

| COMMERCIAL BOAT RISK MITIGATION CHECKLIST: |
| --- |
| ❑ Coordinate with the authorities to help communicate event location and schedule details to local boat operators |
| ❑ Establish alternate routes for boat traffic that are communicated in advance of the event and clearly marked with buoys and signage |
| ❑ Create a defined perimeter around the swim course to insulate swimmers. This perimeter may be comprised of equipment, vessels, and/or swim safety personnel lined up to help protect swimmers |
| ❑ Ensure all race boats and asserts are properly marked and easily identifiable as part of the event support team |

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